BIG MART SALES PREDICTION USING MACHINE LEARNING

Artificial Intelligence Internship Project Presentation

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Organisation: Naviotech Solution • 2026

Introduction

Sales prediction is the process of estimating future product sales using historical data, market trends, and statistical models. It empowers retailers to anticipate demand and align supply accordingly.

Machine learning brings precision to business analytics by uncovering hidden patterns within large datasets — enabling proactive, data-driven decision-making across the retail value chain.

Predictive analytics has become indispensable in the retail industry, where accurate forecasts directly influence profitability, customer satisfaction, and competitive advantage.

Demand Estimation

Forecasting future product demand with statistical modelling

Inventory Optimisation

Aligning stock levels with predicted consumer demand

Strategic Planning

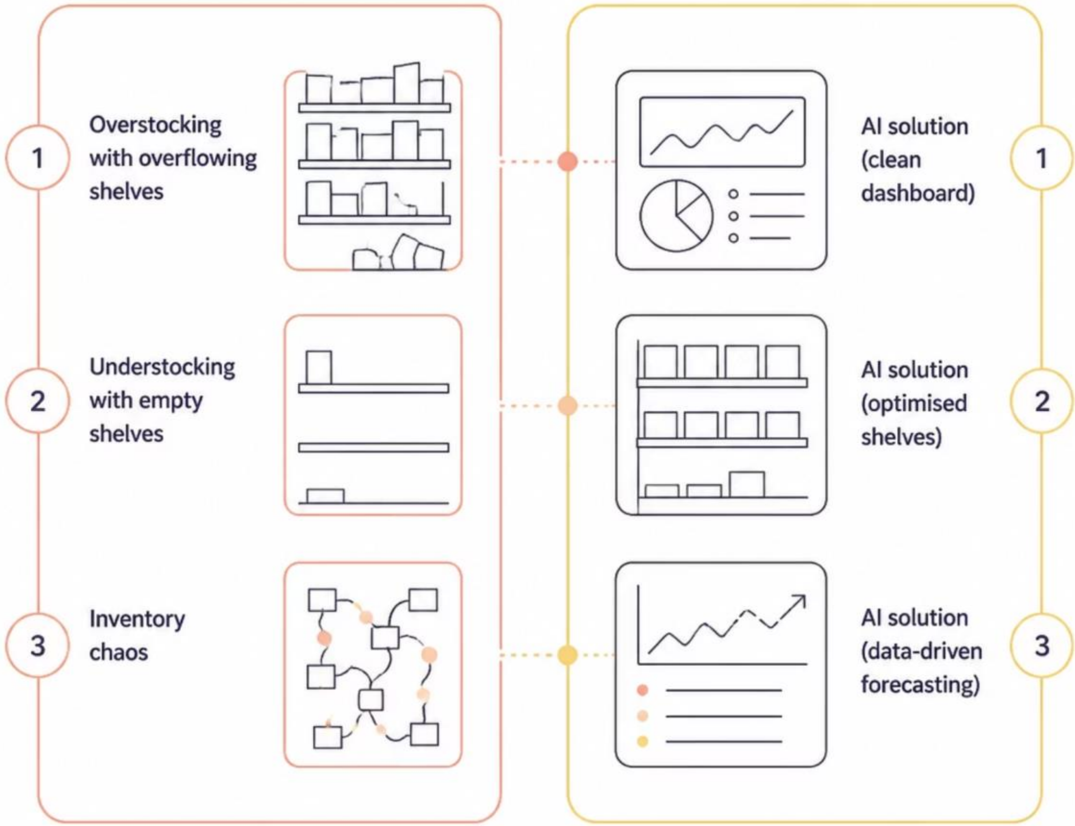
Supporting pricing, promotion, and supply chain decisions

The Problem Statement

Retailers face persistent challenges in inventory management that erode margins and customer trust. Two issues dominate:

- **Overstocking** — Excess inventory ties up capital, increases warehousing costs, and leads to spoilage or obsolescence.
- **Understocking** — Insufficient stock triggers lost sales, customer dissatisfaction, and brand erosion.

Traditional forecasting methods rely on intuition and historical averages — approaches that fail to capture the complexity of modern retail dynamics.



Project Objectives

Predict Product Sales

Build an accurate ML model to forecast outlet-level sales for diverse products.

Improve Inventory Planning

Enable data-backed stock decisions that reduce waste and prevent shortages.

Support Business Decisions

Deliver actionable insights for pricing, promotions, and supply chain strategy.

Apply ML Techniques

Leverage supervised learning and feature engineering on real-world retail data.

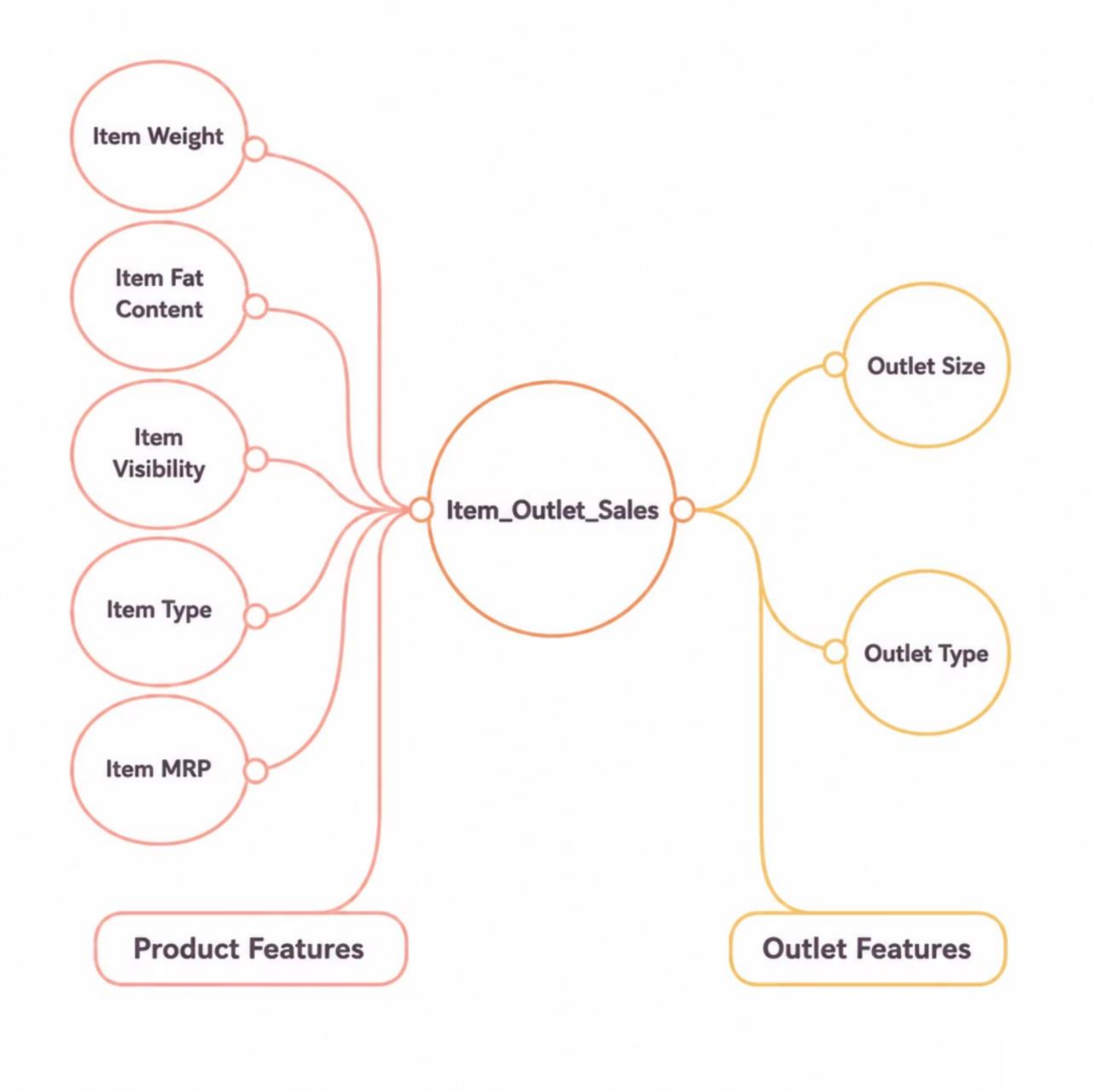
Analyse Retail Data

Explore and interpret Big Mart's historical sales dataset to extract meaningful patterns.

Dataset Description

The Big Mart dataset comprises **8,523 product records** across **1,559 products** and **10 retail outlets**. It is split into two files for model training and evaluation.
Target Variable:

Item_Outlet_Sales — the continuous value our model learns to predict.



Technology Stack

A robust Python-centric ecosystem powering the end-to-end machine learning pipeline — from data wrangling to model deployment.

Python

Core programming language for the entire ML workflow

Pandas

Data manipulation and structured analysis

NumPy

Numerical computing and array operations

Matplotlib

Data visualisation and exploratory plotting

Seaborn

Statistical graphics and correlation heatmaps

Scikit-Learn

Preprocessing, model selection, and evaluation

XGBoost

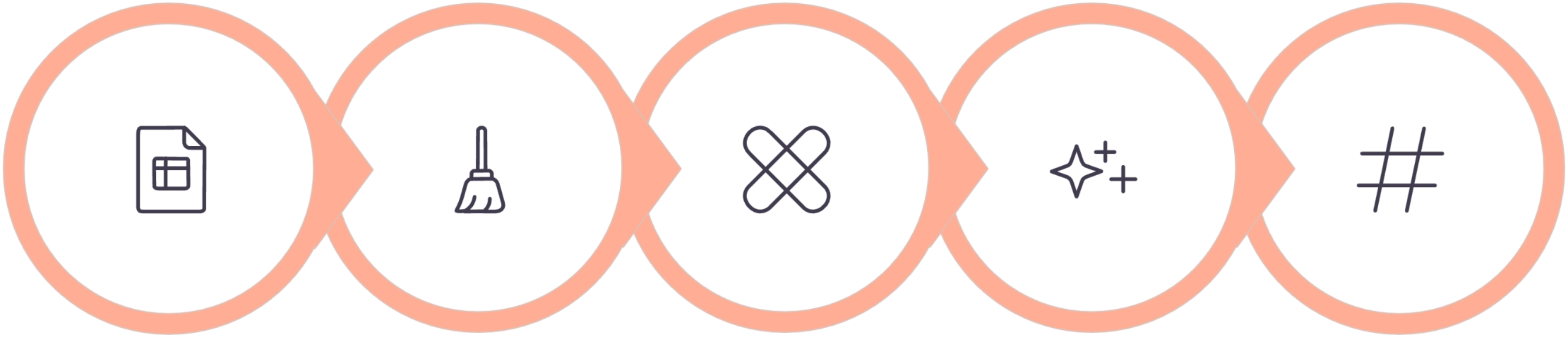
Gradient-boosted regression model for prediction

Google Colab

Cloud-based notebook development environment

Data Preprocessing Pipeline

Raw data is rarely model-ready. A structured preprocessing pipeline ensures data quality, consistency, and predictive power before training begins.



Each stage transforms raw retail records into a clean, encoded, and feature-rich dataset ready for supervised learning.

System Architecture

The end-to-end architecture follows a linear data-to-insight pipeline — from raw Big Mart records through preprocessing and modelling to actionable sales predictions.

01

Dataset

Load Train.csv and Test.csv into Pandas DataFrames for processing

02

Data Preprocessing

Clean, transform, and engineer features from raw retail data

03

Missing Value Handling

Impute gaps using median and mode strategies for robustness

04

Label Encoding

Convert categorical features into numeric representations

05

Train-Test Split

Partition the training data for model validation

06

XGBoost Regressor

Train the gradient-boosted ensemble on processed features

07

Prediction

Generate sales forecasts for the test dataset

08

Performance Evaluation

Assess accuracy using R^2 score and residual analysis

XGBoost Regressor

XGBoost (eXtreme Gradient Boosting) is an optimised distributed gradient-boosting library designed for speed and performance. It builds an ensemble of weak learners sequentially, each correcting errors from its predecessor.

Why XGBoost? It consistently outperforms other regression algorithms on tabular data, making it the natural choice for Big Mart's structured dataset.

1

High Accuracy

Ensemble learning captures complex non-linear patterns

2

Efficient Training

Parallelised tree construction enables rapid convergence

3

Built-in Regularisation

L1 and L2 penalties prevent overfitting on training data

4

Handling Missing Data

Natively learns default directions for sparse features

Implementation Workflow

Load Dataset

Import Big Mart CSV files into Pandas

Explore Data

Uncover distributions, correlations, and outliers

Handle Missing Values

Impute Item_Weight and Outlet_Size gaps

Label Encoding

Convert all categorical columns to numeric

Train XGBoost

Fit the regressor on the training split

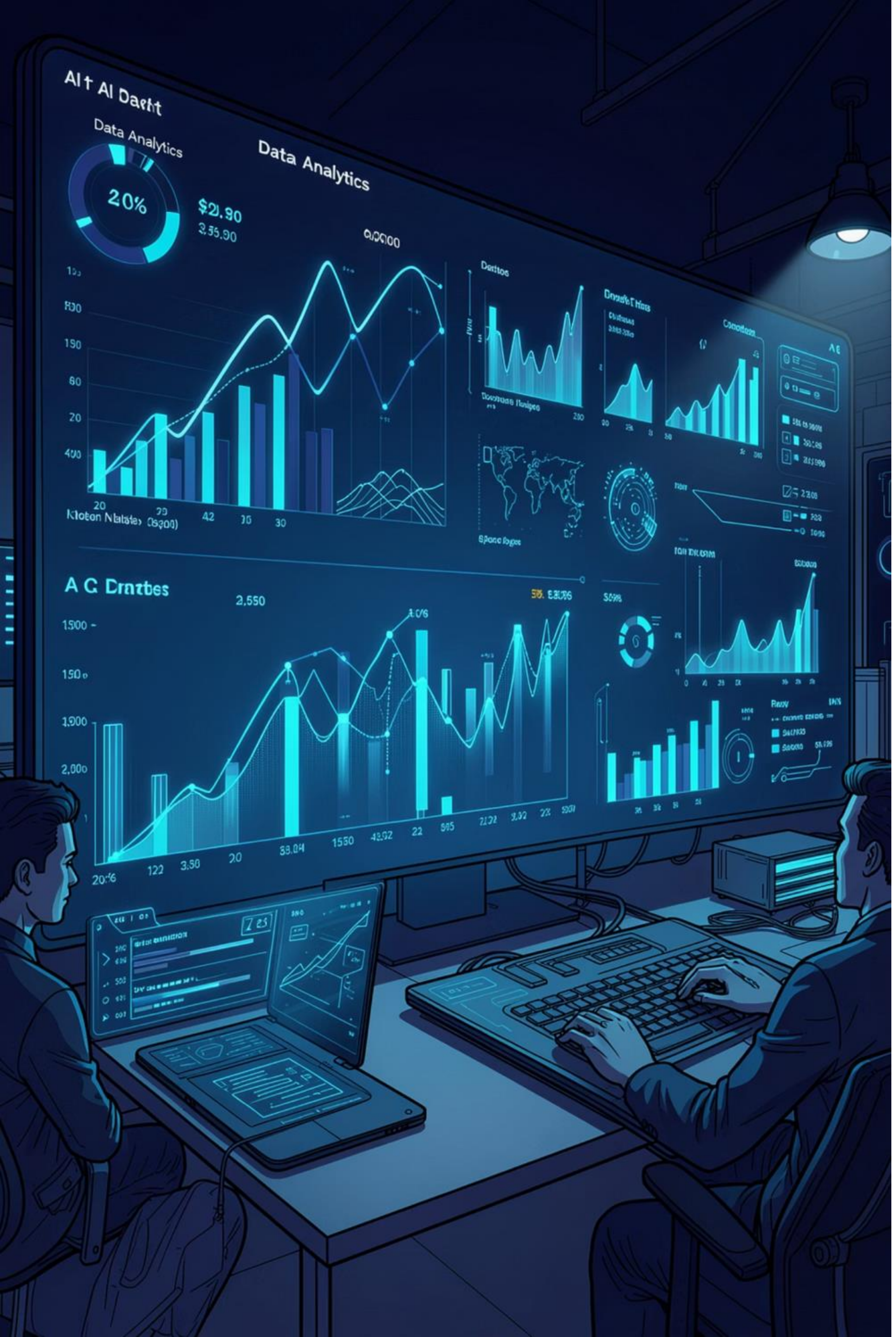
Generate Predictions

Forecast sales for the test dataset

Evaluate Results

Measure performance with R^2 and RMSE

Each stage was executed iteratively in Google Colab, with visual validation at every checkpoint to ensure pipeline integrity.



AI INTERNSHIP PROJECT

NAVIOTECH SOLUTION

Big Mart Sales Prediction Using Machine Learning

A Machine Learning-driven approach to retail sales forecasting using the XGBoost Regressor, presented as part of the Artificial Intelligence Internship Programme at Naviotech Solution.

Presenter

Vinay Kumar OGRE

Organisation

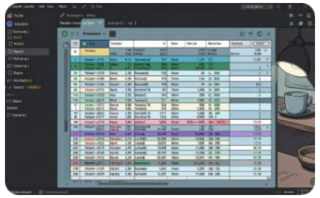
Naviotech Solution

Year

2026

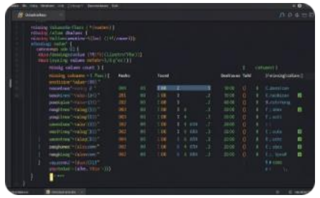
Output Screenshots

Key stages of the model pipeline captured as dashboard outputs, illustrating data ingestion, preprocessing, encoding, and model training.



Dataset Loading Output

The Big Mart dataset is loaded into a structured DataFrame, displaying product and outlet-level features across 8,523 records.



Missing Values Output

Null value analysis identifies gaps in Item_Weight and Outlet_Size, guiding the imputation strategy for data integrity.

Label Encoding Output

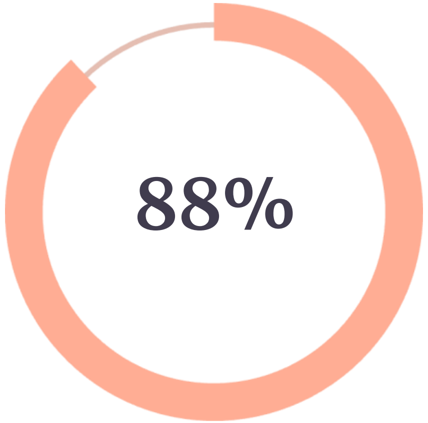
Categorical variables such as Item_Fat_Content and Outlet_Type are encoded into numerical labels for model compatibility.

Model Training Output

The XGBoost Regressor is trained on processed features, with iterative boosting rounds converging towards optimal performance.

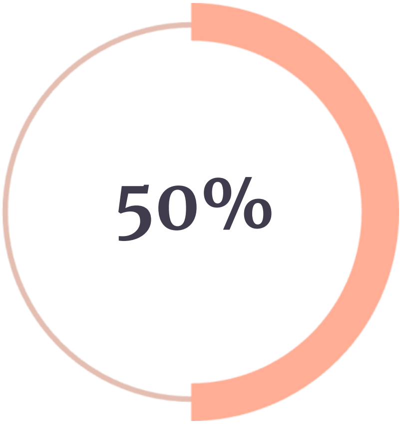
Model Performance & Results

The XGBoost Regressor was evaluated on both training and testing datasets to assess generalisation capability and predictive accuracy.



Training R^2 Score

0.8762



Testing R^2 Score

0.5017

The model achieves a strong training R^2 score of 0.8762, indicating it learned the training patterns well. The testing R^2 score of 0.5017 reflects moderate generalisation to unseen data.

Training Accuracy

High fit on training data with R^2 = 0.8762, capturing complex nonlinear relationships.

Testing Accuracy

Test R^2 = 0.5017 suggests potential overfitting, indicating room for regularisation tuning.

Advantages & Future Scope

Better Sales Forecasting

Accurate predictions of product-level sales across retail outlets.

Improved Inventory Management

Optimised stock levels reduce wastage and shortage risks.

Business Growth

Data-backed insights unlock expansion and profitability.

1. **Web Application Deployment**

   Deploy as an interactive web app using Flask or Streamlit for stakeholder access.

2. **Cloud Integration**

   Scale the model on AWS or Azure with serverless inference endpoints.

3. **Real-Time Prediction**

   Stream live sales data for instant forecasting and dynamic pricing.

4. **Advanced AI Models**

   Explore deep learning ensembles and hyperparameter tuning for higher accuracy.

Conclusion & Thank You

The Big Mart Sales Prediction project successfully demonstrates the application of Machine Learning for retail sales forecasting. The **XGBoost Regressor** model delivered a training R^2 score of 0.8762, capturing complex relationships across product and outlet features, while the testing R^2 of 0.5017 highlighted areas for further regularisation and feature engineering.

This project establishes a scalable foundation for data-driven retail decision-making, inventory optimisation, and business intelligence.

Thank You

Questions & Answers